

PRIMARY EAR AND HEARING CARE

TRAINER'S HANDBOOK



World Health Organization

Primary ear and hearing care: trainer's handbook

ISBN 978-92-4-006958-9 (electronic version)
ISBN 978-92-4-006959-6 (print version)

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Cataloguing-in-Publication (CIP) data. CIP data are available at **http://apps.who.int/iris**.

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Section 1

INTRODUCTION

Welcome to the *Trainer's handbook* which accompanies the *Primary ear and hearing care training manual*. This handbook is intended for use by those involved in training health workers or in coordinating programmes for delivering ear and hearing care.

Why do we need training in primary ear and hearing care?

Over 6% of the global population is affected by disabling hearing loss;[1] over 80% of these people live in low- and middle-income countries, most of which lack the specialist human or other resources to prevent, diagnose, or treat such disorders (1). For example, a report from 2021 revealed that most countries in the WHO African Region had fewer than one audiologist, or one ear nose and throat (ENT) specialist, available per 1 million people (2).

Training specialist health workers, such as audiologists or ENT specialists, is costly, time-consuming, and may reinforce a hospital-based model of care that can fail to reach disadvantaged or remote populations. Given that ear and hearing problems are some of the most common conditions experienced by populations, it is important that services for addressing these should be available as part of primary health care (3). One solution is task-shifting and task-sharing, through the training of non-specialist health workers such as primary care doctors or health workers.[2]

The Primary ear and hearing care training manual is a resource for training non-specialist workers in the recognition and initial management of ear disease and hearing loss. It replaces the previous set of manuals released by the WHO in 2006.

In settings where the training of lay workers, such as health volunteers or members of the public, is proposed, the WHO's Basic ear and hearing care resource can be used (4). This resource can enable trainees to raise awareness in their communities on the prevention and early identification of common ear and hearing problems and sensitize them to the needs of people with hearing loss.

What is the evidence to support the WHO approach to training primary health workers in ear and hearing care?

A 2019 review of published evidence identified that community health workers (CHWs) were successfully involved in screening for ear and hearing disorders, raising community awareness, and delivering basic levels of care (5). However, in terms of training, concerns were raised that previous studies in this area had tended to evaluate learning through an assessment of knowledge gained which was measured using only a multiple-choice questionnaire completed

[1] Disabling hearing loss refers to moderate or higher severity of hearing loss (above 35 dBHL) in the better hearing ear.
[2] Health worker is an umbrella term for those working within their own community in a role of health promotion, disease prevention and identification, and delivery of care.

immediately at the end of training. There was no published evidence on assessment of clinical competency (as opposed to knowledge gained), nor of long-term retention and application of skills.

In 2021, a study was undertaken in Uganda of CHWs who were trained using a field test version of the current manual (6). At the end of their training course, their learning was assessed through the use of an Objective Structured Clinical Examination (OSCE) – an approach to testing shown to reliably assess clinical competencies in other contexts (7). During the following six months the CHWs were provided with scheduled and ad-hoc peer-to-peer educational support, delivered either in person or through digital platforms. When OSCEs were repeated at the end of this period, the trained CHWs showed retention of skills and performance. The content of the manual and testing methods used in Uganda have since undergone additional field testing in Fiji, India and Zambia.

What is in this Trainer's handbook?

The Trainer's handbook contains a guide to structuring the training course and a suggested timetable – although each training course should be adapted to local needs. An overview of the Training manual is also provided, as well as its accompanying resources, the learning objectives for each module, a list of the requirements for each module, and tips for teaching. In addition the handbook includes a guide to conducting assessments, and an example of an end-of-course assessment carried out using the OSCE method.

Both the handbook and manual are intended to be valuable training resources. Additional support, if required, is available by contacting the WHO offices at whf@who.int.

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Section 2

PREPARING AND STRUCTURING THE TRAINING COURSE

When planning and preparing for the training course, it is recommended that all trainers first read the Trainer's handbook and the Training manual in their entirety. The manual consists of an introduction to the importance of hearing, followed by nine modules which address the anatomy of the ear, identification of common ear and hearing problems, management at primary level of care, and rehabilitation. It ends with information on WHO's actions for promotion of ear and hearing care. Key aspects for preparation of the training course are outlined below.

Determining the training objectives and scope

The objectives and scope of the training will depend on the local needs and services to be delivered by the trained health workers or doctors. It may be considered appropriate to deliver the entire training course, or, in some settings, to select relevant parts only. For example, if services already exist for screening for hearing loss in children, this part could be omitted from the training.

Adapting the Training manual to local needs

The Training manual should be adapted to best meet the training objectives.

It is recommended that any adaptation should:

- give national or local data on hearing loss;
- focus on the ear and hearing problems most commonly experienced in the target population;
- if required, modify recommendations made in the manual (regarding medicines given for treatment of ear conditions) to suit the local context and regulations (according to who is permitted to prescribe various medicines mentioned in the manual);
- add photographs or images most relevant to the local context; and
- identify the most relevant modules for training, if required. For example, if the training focuses on ear diseases alone, trainers can choose to use modules 1–5, and 9. If the purpose of training is to train workers in prevention and identification of hearing loss, modules 1, 7–10 can be selected.

Accompanying resources

WHO has developed several accompanying resources that can facilitate the delivery of the training course as well as the subsequent activities of the health workers or doctors. These include:

- Presentation slides which illustrate the theoretical aspects of the training. The slides can be used when delivering the training and can be adapted to suit the local context and need. The slides are available on request by emailing: whf@who.int.

- A set of videos prepared by WHO to accompany the Training manual. The videos can facilitate the teaching of practical skills such as otoscopy and ear washout. They are available on request by emailing whf@who.int.
- Community resources for raising awareness. Six accompanying flyers can be downloaded from the WHO website. Weblinks are provided in the Training manual as web annexes. These community resources are intended as information flyers which health workers or doctors can share with their patients or with the general public as indicated. Trainers should print copies of these resources and share them during the training programme so that trainees can familiarize themselves with the use of these resources. The trainees should be encouraged to distribute these resources, where appropriate, when working in the field.
- The hearWHO and hearWHOpro apps: hearWHO is a cost-free WHO app that can be used by anyone to assess and track their own hearing. The hearWHOpro app is intended for use by health workers to check the hearing of people in their communities. Multiple patient records can be saved and downloaded. Both apps are available from Google playstore and Appstore by searching "hearWHO", and can be used for demonstration purposes. Trainees with access to devices with android or iOS platforms can be asked to download these apps during the training.

Requirements for conducting the course

- Experts in the field of ear and hearing care to help deliver the training and to undertake assessments.
- A suitable venue for conducting the course. Ideally this should be located close to a health facility so that trainees can benefit from practical learning with patients from the facility. The venue should also have adequate space to conduct classroom training and the group activities outlined in the course.
- A computer and projector for delivering lectures.
- Sufficient printed copies of the Training manual for all trainees. Ideally each trainee should have their own printed copy of the manual for use during the training and as an information resource following completion of their training.
- Equipment required for each specific module can be found on page 10 . The list is also provided within the Training manual at the start of each module.
- Patients for demonstration of certain clinical findings, wherever possible. This is specified for each module. It is important to seek full permission/consent from the patients prior to their participation.
- Financial resources to cover for potential lunch, transport and accommodation allowances, according to local practice.

Timetable

When all modules are included, the training is expected to take four days in total – three days for the training, and an additional day for assessment. However, if necessary, it is possible to schedule longer training days so that the modules and assessment are completed within three days. Another alternative is for the assessment day to take place at a later date if more convenient. The number of hours required for each module are suggested below in Table 1.

Table 1. Estimated time requirement per module

Topic	Suggested number of hours needed	Comment
"Icebreaker" and introduction	1 hour	An "ice breaker" should be included to allow trainees to familiarize themselves with each other and the trainers
Module 1: Mechanism of hearing and anatomy of the ear	1 hour	–
Module 2: Assessing ear and hearing problems	2 hours	Includes 2 practical trainings and 2 activities
Module 3: The outer ear: diagnose, treat, refer	3 hours	Includes 3 practical trainings and 3 activities
Module 4: Infections of the middle ear: diagnose, treat, refer	3 hours	Includes 5 activities
Module 5: Other problems of the middle ear: diagnose, treat, refer	1 hour	–
Module 6: Hearing loss: grades, causes and prevention	1 hour	Includes 1 activity
Module 7: Identifying hearing loss in children and adults	3 hours	Includes 4 practical trainings and 2 activities
Module 8: Rehabilitation of hearing loss, including hearing aids and cochlear implants	1 hour	Includes 2 activities
Module 9: The role of health workers and doctors in ear and hearing care	0.5 hour	–
Module 10: What does the World Health Organization do?	0.5 hour	–
Test: In groups of 5 or 6	4-5 hours	–

Discussion, activities and practicals

Discussion points: Each module includes discussion points. These are questions that can be posed to the trainees as a means for stimulating discussion. They can help to break the routine of classroom teaching and make the learning experience interactive. Trainers should modify these, as needed, to suit the local context and training schedule.

Activities: Each module includes activities that aim to help trainees practice the skills they are learning. All activities should be reviewed prior to delivering the module and availability of the required equipment/patients/material for the activity ensured.

Practicals: Nine practical trainings are included in the modules. Each practical training describes a specific clinical skill that trainees will be expected to practice once they are working in the field. It is important to ensure that appropriate time and opportunities are provided for all trainees to learn and practice these skills.

Assessment

Each module is accompanied by a pre-test and a post-test. Trainees should be asked to take the pre-test at the start of the module. Their responses can be used as a tool for stimulating discussion on the topics to be covered in that module and to gauge the extent of their pre-existing knowledge. Trainees should be asked to take the post-test at the end of each module. This provides a means for testing their immediate gains in knowledge.

The final assessment should be undertaken, preferably with an OSCE to assess clinical skills and knowledge retention. Note that OSCEs are the preferred approach for end-of-training assessment.

Feedback and evaluation

Post-training evaluation and feedback of the training programme can be very helpful. Wherever possible this should be sought in a structured manner. Any feedback regarding the Training manual itself can be conveyed to the World Health Organization by emailing: whf@who.int. This will help inform improvements to the next version of the manual.

Ongoing support

It is strongly recommended that plans are made for the ongoing support of trainees after completion of their training course. This could include a short refresher course on knowledge and skills, and/or the creation of an electronic chat group for discussion and referral of cases (with appropriate safeguards to protect confidential patient information).

Section 3

LEARNING OBJECTIVES, TEACHING TIPS AND EQUIPMENT REQUIRED FOR EACH MODULE

The Training manual comprises 10 modules. The learning objectives, equipment required, and new terms used in each module are listed at the start of each module. For ease of reference, these are compiled below. In addition, teaching tips are provided for trainers to facilitate the training sessions.

Introduction

Teaching tips:

- Start the training by a round of introductions, which can be facilitated by use of an "icebreaker".
- To introduce the trainees to the subject, start by asking if they know or have met anyone with hearing loss and what their experience was of interacting with that person or persons.
- Ask trainees what solutions they may have seen people in their community using – for example, hearing aids, implants, sign language etc. Ask trainees what they think the causes of hearing loss may be.
- Use the information provided in the introduction about the prevalence of hearing loss to explain how common hearing loss is.
- Stress the fact that most common ear diseases can be treated.
- Emphasize that rehabilitation of hearing loss is most effective when it is identified early.
- Discuss the role they can play in making sure that people with ear diseases and hearing loss receive the care they need.

Module 1:

Mechanism of hearing and anatomy of the ear

On completion of this module, trainees should be able to:

- Recognize and name parts of the ear.
- Understand the structure of the ear.
- Explain the functions of different parts of the ear.
- Explain the path through which sound travels in the ear.
- Explain how people hear.

Teaching tips

- First ask trainees about the sounds they like and dislike, and how they think these sounds are heard.
- If possible, use a model to explain the structure of the ear. Alternatively, print out the diagrams included in the Training manual.
- In order to better explain the mechanism of hearing, an animation such as this animation can be used. Trainers can use any simple and

suitable animation that is freely available in the language required.

What you will require

No equipment is required for this module.

Module 2:

Assessing ear and hearing problems

By the end of this module, trainees should be able to recognize:

- Common ear complaints and what they could mean.
- How to take a case history of ear and hearing problems.
- How to examine the ear with the use of an otoscope and what to expect when doing this.

Teaching tips

» Start by asking trainees what ear problems and complaints they have commonly encountered in the community. Ask them what they think may be the causes of these problems.

» Explain that assessment of ear and hearing problems has three parts: history-taking, ear examination and hearing assessment (as explained in Module 7).

» Explain history-taking with the help of examples that trainees may commonly see in their community, e.g. ear pain in a child; a child unable to hear or speak, or with poor performance in school; an adult with hearing loss etc.

» First explain about conducting an examination of the ear, then follow with a demonstration. Allow trainees time to practice an examination on each other, including otoscopy.

» Allow adequate time for trainees to perform the activities included in the module.

» Make print-outs of the activity response sheet if required.

What you will require

- Pen and paper
- Otoscope with speculum
- Community resource 1: when to suspect hearing loss (printed copies).

Module 3:

The outer ear: diagnose, treat, refer

In this module trainees will learn about:

- Common diseases of the outer ear including the pinna and ear canal.
- How to remove ear discharge and foreign bodies from the ear by washout.
- Actions that can be taken to treat these problems in the community.
- Treatment which may need to be given by specialists for certain outer ear conditions.
- When to refer patients with outer ear problems to a doctor/specialist.

Teaching tips

» Start by asking the trainees to identify common parts of the outer ear, using each other as examples, and their manual, as part of Activity 3.1.

» Explain the common problems of the outer ear, with help of some patients, if possible.

» Explain practical skills to trainees such as ear washout and the insertion of ear drops. Follow with a demonstration on an individual.

» Ask trainees to practice these skills on each other.

» Demonstrate to trainees some of the common problems of the outer ear, such as wax, otitis externa, ear discharge etc, if possible using patients so that trainees can see a real-life presentation of these conditions.

» Emphasize the situations where trainees should not perform an ear washout.

» Stress those conditions where an urgent referral to a medical specialist is required (indicated in red within the manual).

What you will require

- Otoscope with speculum
- Clean water (boiled if necessary)
- 20 ml syringe (without needle)
- Kidney dish or other bowl
- Tissues
- Ear drops
- Cotton and wooden stick applicator
- People with some of the common ear problems (for demonstration)

Module 4:

Infections of the middle ear: diagnose, treat, refer

By the end of this module the trainee should be able to:

- Recognize common diseases of the middle ear.
- Make and use a dry mop safely to clean the ear canal.
- Make and use a wick to clean discharge from the ear canal.
- Treat acute otitis media.
- Guide patients on how to manage a discharging ear.
- Refer patients, when required.

Teaching tips

» Start by revising the structure and function of the middle ear. Ask trainees to label parts of the middle ear in their manual as part of Activity 4.1.

» Explain the common problems of the middle ear, if possible by using patients as examples.

» Revise the use of an otoscope as well as dry mopping and wicking of the ear. Ask trainees to practice these skills on each other.

» Demonstrate to trainees the clinical findings of common problems such as acute otitis media, tympanic membrane perforation and mastoiditis, if possible by using patients with these conditions.

» Emphasize that giving patients proper instructions regarding keeping ears dry is as important as giving proper medicines.

» Stress those conditions where an urgent referral to a medical specialist is required (indicated in red within the manual).

» Inform trainees that the community resources included in this manual are to be used as informational flyers for people who need them. Print out copies to familiarize trainees with these. Encourage trainees to use these resources when they have returned to the field.

What you will require

- Otoscope with speculum
- Wooden stick applicator
- Cotton
- Tissues
- People with common middle ear problems (for demonstration)
- Community resource 2: Information for patients with discharging ears (printed copies).

Module 5:

Other problems of the middle ear: diagnose, treat, refer

By the end of this module the trainee should be able to:

- Recognize the signs of glue ear and cholesteatoma.
- Guide patients with glue ear and refer if required.
- Guide patients on the importance of addressing cholesteatoma as early as possible.

Teaching tips

» Explain that glue ear is very common in young children and can often remain undetected.

» Emphasize that cholesteatoma can lead to life threatening complications.

» If possible, try to make available patients who have glue ear or a cholesteatoma, so that trainees can experience a real-life presentation of these common ear problems.

» Stress those conditions where an urgent referral to a medical specialist is required (indicated in red within the manual).

What you will require

- Otoscope with speculum
- People with ear problems, as indicated (for demonstration).

Module 6:

Hearing loss: grades, causes and prevention

By the end of this module the trainee should be able to:

- Discuss the common causes of hearing loss and how these may be prevented.

Teaching tips

» Start by asking the trainees to block their own ears with earplugs or cotton wool while carrying on a conversation. Ask them how well they hear and how it makes them feel.

» Speak to them about the common causes of hearing loss at different stages of life: before birth; at birth; during childhood; in adults; in old age; at any stage of life.

» With each cause of hearing loss, discuss if there are ways and means to prevent it and what these are.

» Ask the trainees to list the different ways in which they can promote the prevention of hearing loss in their community.

What you will require

No equipment is required for this module.

Module 7:

Identifying hearing loss in children and adults

By the end of this module the trainee should be able to:

- List the common milestones for hearing in babies and children.
- Suspect hearing loss in children and adults.
- Perform different tests (without equipment) depending upon the age of the individual.
- Use hearWHOpro or similar methods for testing.
- Understand when a person should be referred for further testing.
- Know about tests that are conducted at a medical facility.

Teaching tips

» Start by asking the trainees if/how they have noticed the development of hearing and speech milestones in a baby. Ask them about their experiences with babies and if they know at what stage babies start to turn their heads towards sounds or start to babble.

» Explain that speech and hearing are linked and that a delay in speech development in children is often an indication of hearing loss.

» Demonstrate clearly each of the tests of hearing – i.e. distraction test, voice test, and whispered voice test. Ask trainees to practice each of these.

» Demonstrate the use of the hearWHO and hearWHOpro apps. To do this, first make sure that these apps are downloaded onto a smartphone or tablet and that you have familiarized yourself with them.

» Explain the tests that can be conducted in a baby/child/adult to test their hearing in a medical facility. Stress the fact that hearing can be tested at any age with the proper equipment. If possible, arrange for these tests to be demonstrated to the trainees.

» Stress those conditions where an urgent referral to a medical specialist` is required (indicated in red within the manual).

What you will require

- Wooden blocks or similar toy that makes no sound
- A towel or sheet
- A rattle
- One or more smartphones with hearWHO and hearWHOpro downloaded
- Headphones or earphones (if possible, use with noise cancellation)
- Community resource 3: Speech and hearing landmarks in children (printed copies).

Module 8:

Rehabilitation of hearing loss

By the end of this module the trainee should be able to:

- Explain why early identification and intervention for hearing loss is important.
- List the different ways and means of rehabilitation for persons with hearing loss.
- Instruct a person on how to care for their hearing aid.
- Understand why the use of sign language can benefit all people with hearing loss.
- Inform others about the importance of loops and FM systems in schools.
- Suggest means for communication access to people with hearing loss, e.g. alarm signallers, captioning etc.

Teaching tips

» Start by asking the trainees what they know about hearing rehabilitation. Have they seen anyone using a hearing aid? Or a cochlear implant? Have they met anyone who uses sign language?

» Ask trainees why they think rehabilitation is important both for children and for adults.

» If possible, invite a parent of a deaf child to join the group and share their experiences of early identification of hearing loss and its importance.

» Show trainees a hearing aid; explain how it works and how it should be taken care of. If possible, arrange for a person who uses a hearing aid to join the group and share their experiences.

» Demonstrate to trainees how to care for hearing aids. Do this in a manner so that they will be able explain the procedure to users of hearing aids. Community resource 4 should be used for this.

» Show trainees a model of a cochlear implant and if possible, arrange for a person who uses a cochlear implant to join the discussion and share their experiences.

» Emphasize that the use of hearing aids or implants, especially in children, must be accompanied by appropriate therapy.

» Emphasize that the use of sign language can benefit all those who are deaf, including those who cannot use and benefit from hearing devices. Try to introduce the trainees to a person who uses sign language so they can learn more about this method of communication.

» Encourage trainees to advocate with school and government authorities for the provision of services such as loop systems in schools and captioning on television.

What you will require

- Hearing aid with battery
- Model of cochlear implant (if possible)
- Person/s using a hearing aid, or cochlear implant, parent of deaf child, user of sign language
- Community resource 4: Tips for hearing aid users (printed copies)
- Sealed box and dry material for storing hearing aids .

Module 9:

The role of health workers and doctors in ear and hearing care

By the end of this module the trainee should:

- Know what role they can play in the prevention, early identification, and management of hearing loss and ear diseases.
- Be able to counsel people regarding the “Dos” and “Don’ts” of ear and hearing care.
- Understand simple steps that can protect ears against loud sounds including music.

Teaching tips

» Start by asking trainees what they consider their role is in ear and hearing care.

» Ask trainees about noise in their community and if people commonly listen to loud sounds through headphones/earphones.

» At the end of the module, ask trainees what they plan to do to promote ear and hearing care once they have returned to their community.

What you will require

Community resource 5: Tips for healthy ears (printed copies).

Community resource 6: Tips for safe listening (printed copies).

Module 10:

What does the World Health Organization do?

This module introduces the trainee to the work being carried out by WHO. If not considered relevant this can be excluded.

Additional module: Local context

A separate module on local context can be included in the training. This should focus on:

- prevalence of hearing loss in the region/ country/area;
- the ear problems most commonly encountered;
- availability of ear and hearing care services in the country/area; and
- referral centres and other local infrastructure.

Section 4

ASSESSMENT

Assessment included in this Trainer's handbook consists of a general written test and four themed OSCE stations (a station refers to one problem as part of the OSCE) to test practical skills and application of knowledge[3]:

- **Station 1: Written test (optional)**
- **Station 2: OSCE problem 1: Examination**
- **Station 3: OSCE problem 2: Skills assessment**
- **Station 4: OSCE problem 3: Clinical scenarios**
- **Station 5: OSCE problem 4: Hearing test: e.g. Distraction test of hearing**

Each of the assessment stations including the written test, requires 15 minutes. The number of stations to be tested will depend on which tests are selected for inclusion. Inclusion of the written test is optional.

It is important to note that:

- OSCE stations 1, 2 and 4, need to have an observer-examiner present to assess and score the candidate on their performance. Observer -examiners should have relevant experience of ear and hearing care.
- If possible, it is preferable for OSCE station 3 to also have an observer-examiner who can then ask the questions and score directly onto the mark sheet. If not possible, the question sheet for OSCE station 3 and the answer sheet can both be printed and left for candidates to complete themselves.
- The written test does not need an observer-examiner.
- It is important to have a general helper to assist with the exam. By ringing a bell or other sound device every 15 minutes, the helper can alert the candidates that the station time is ended and that they must rotate. The helper can also be available to support candidates and collect answer sheets at stations that do not have an examiner. The helper does not need to have any expertise in ear and hearing care.

An example set-up of examination stations is described below in Table 2. Each round is 15 minutes. The example presented includes five testing stations (four OSCEs and one written test), but can be adapted to the number of stations you have. Candidates can be tested in groups, with the size of the batch matching the number of candidates. In the example there are five stations, so five candidates are tested, and the total time taken for testing is 15 x 5 minutes = 75 minutes. A break of 5 or 10 minutes can be given for the examiners to rest before starting the next round of testing.

[3] Khan, Kamran Z.; Ramachandran, Sankaranarayanan; Gaunt, Kathryn; Pushkar, Piyush (2013). "The Objective Structured Clinical Examination (OSCE): AMEE Guide No. 81. Part I: An historical and theoretical perspective". Medical Teacher. 35 (9): e1437–e1446. doi:10.3109/0142159X.2013.818634

Table 2. An example set-up of examination stations

	Round 1	Round 2	Round 3	Round 4	Round 5
Trainee 1	Written test	OSCE station 1	OSCE station 2	OSCE station 3	OSCE station 4
Trainee 2	OSCE station 1	OSCE station 2	OSCE station 3	OSCE station 4	Written test
Trainee 3	OSCE station 2	OSCE station 3	OSCE station 4	Written test	OSCE station 1
Trainee 4	OSCE station 3	OSCE station 4	Written test	OSCE station 1	OSCE station 2
Trainee 5	OSCE station 4	Written test	OSCE station 1	OSCE station 2	OSCE station 3

The equipment and human resources required for each station are listed in the description for each OSCE station. For the written test, candidates will need a pen or pencil.

Based upon experience, a pass mark of 60% is suggested for each assessment; that is:

- 12/20 points for OSCE 1 or 2.
- 12/20 points for OSCE station 3 if an observer-examiner is present.
- 6/10 points for OSCE 4.
- 12/20 points for the written test.

If a candidate is a borderline fail on one assessment but has passed all the other assessments, they may be allowed to pass if deemed appropriate. Any candidate that fails may be given an opportunity to repeat the test, either on the same day or on a later day, as suitable.

If there is time, it can be helpful to go through the written test with all the candidates who passed the assessment; discussion of the correct answers can aid learning.


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